

# Bayes decision theory

## Introduction

- In this lecture we will cover the following topics:
  - Bayesian decision theory
  - The naïve Bayes assumption
  - Bayes classifier minimizes probability of error
  - Risk minimization
  - Discriminant functions and decision surfaces

## Spam

- It's the early 2000, and spam emails offering you to buy Viagra are flourishing.
- Cannot just filter on keywords.
- Mathematically, we want to find  $P(\text{spam}|\text{content})$ .



**Best Buy Viagra Generic Online**

Viagra 100mg x 100 Pills \$125. Free Pills & Reorder Discount. We accept VISA & E-Check  
Payments. 90,000+ Satisfied Customers!

**Top Selling 100% Quality & Satisfaction guaranteed!**

## Tokenization and learning

- Break email into words  $W_i$
- Discrete events:  $P(W_i) \geq 0$
- Notation
  - $S$ : Spam (illegitimate messages)
  - $H$ : Ham (legitimate messages)
- How to get an expression for  $P(\text{spam}|\text{content})$ ?
- Bayes' rule:
  - $P(S|W_i)$ 
$$= \frac{P(W_i|S)P(S)}{P(W_i)}$$
- where:
  - $P(W_i) = P(W_i|S)P(S) + P(W_i|H)P(H)$

## How to learn from data

- Collect Spam and Ham and learn (estimate)  $P(W_i|S)$  and  $P(W_i|H)$ !
- Example:  $W_i = \text{"Money"}$
- Let us look at the data — pick which word to inspect below.

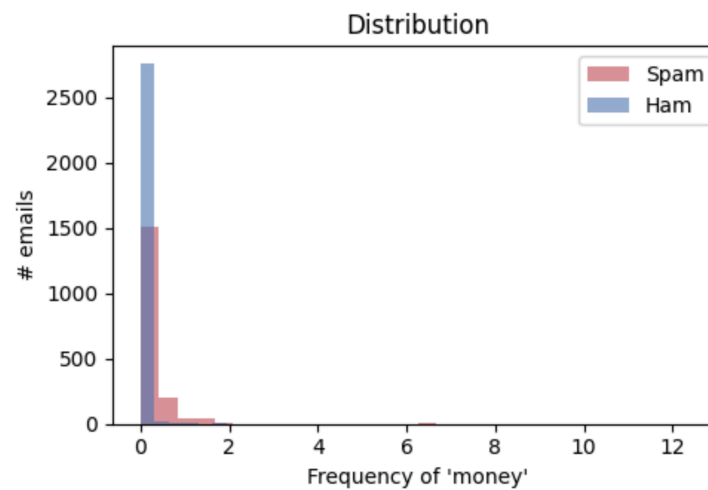
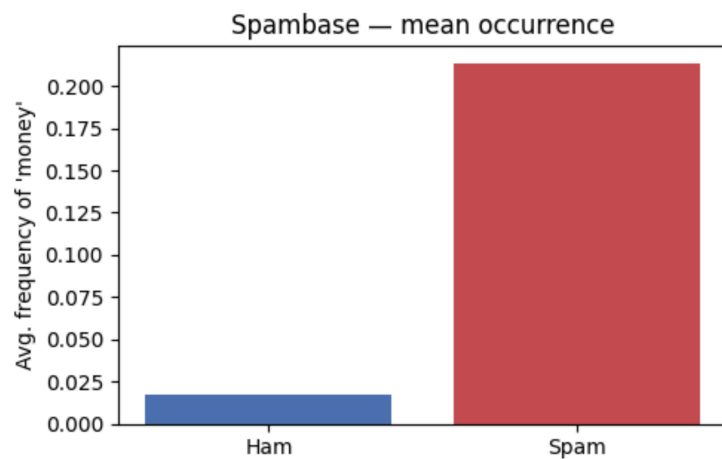




money

george

Spam mean: 0.213 Ham mean: 0.017



## Detection

- Need  $P(W_1 \cap W_2 \cap \dots \cap W_K | S)$
- Difficult!
  - Think about what this term is actually saying.
- Need to simplify.
- Note: modern large language models are modeling something very similar to the expression above.

### 3.1 Unsupervised pre-training

Given an unsupervised corpus of tokens  $\mathcal{U} = \{u_1, \dots, u_n\}$ , we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

where  $k$  is the size of the context window, and the conditional probability  $P$  is modeled using a neural network with parameters  $\Theta$ . These parameters are trained using stochastic gradient descent [51].

#### References:

- Radford et al. — *Improving Language Understanding by Generative Pre-Training*, 2018.
- Bengio et al. — *A Neural Probabilistic Language Model*, 2003.



## Naïve Bayes assumption

- Assume independence!
- $P(A \cap B)$   
 $= P(A)P(B)$
- Then we can compute  
 $P(S|W_1 \cap W_2 \cap \dots$   
 $\cap W_K)$   
:

$$P(S|W_1, \dots, W_K) = \frac{P(S) \prod_i P(W_i|S)}{P(W_1, \dots, W_K)}$$

# Bayes classifier

- Shifting focus a bit.
- How can we think more generally about classification using Bayes' rule?
- First: think continuous random variables  $\rightarrow$
- Let us start with considering two classes  $w_1$  and  $w_2$ .

## Core idea of Bayes classifier

- Given a new sample  $\mathbf{x}$ , how can we decide if we want to classify it as belonging to class  $w_1$ ?
- What about:
  - $P(w_1|\mathbf{x}) > P(w_2|\mathbf{x})$ ?
- else class  $w_2$ .

## Leveraging Bayes' rule

- $P(w_1|\mathbf{x})$  is not straightforward to model.
- Bayes' rule to the rescue:
  - $P(w_1)p(\mathbf{x}|w_1)$   
 $> P(w_2)p(\mathbf{x}|w_2)$
- Much easier to work with  $\rightarrow$

## Example with balanced dataset

- Consider the case where  $P(w_1) = P(w_2) = 0.5$ .
- Draw  $\rightarrow$

## Example with unbalanced dataset

- Consider the case where  $P(w_1) = 0.25$  and  $P(w_2) = 0.75$ .
- Draw  $\rightarrow$

## Example with overlapping Gaussians

- Each tab is a different parameter setting — read the label to see exactly which parameters are in play. Means, standard deviations, and (in the last tab) the prior can all change.
- Click the "**decision boundary**" entry in the legend to toggle it on or off. It starts **off** so the densities stand alone first; turn it on to see where the Bayes classifier would assign each  $x$ .



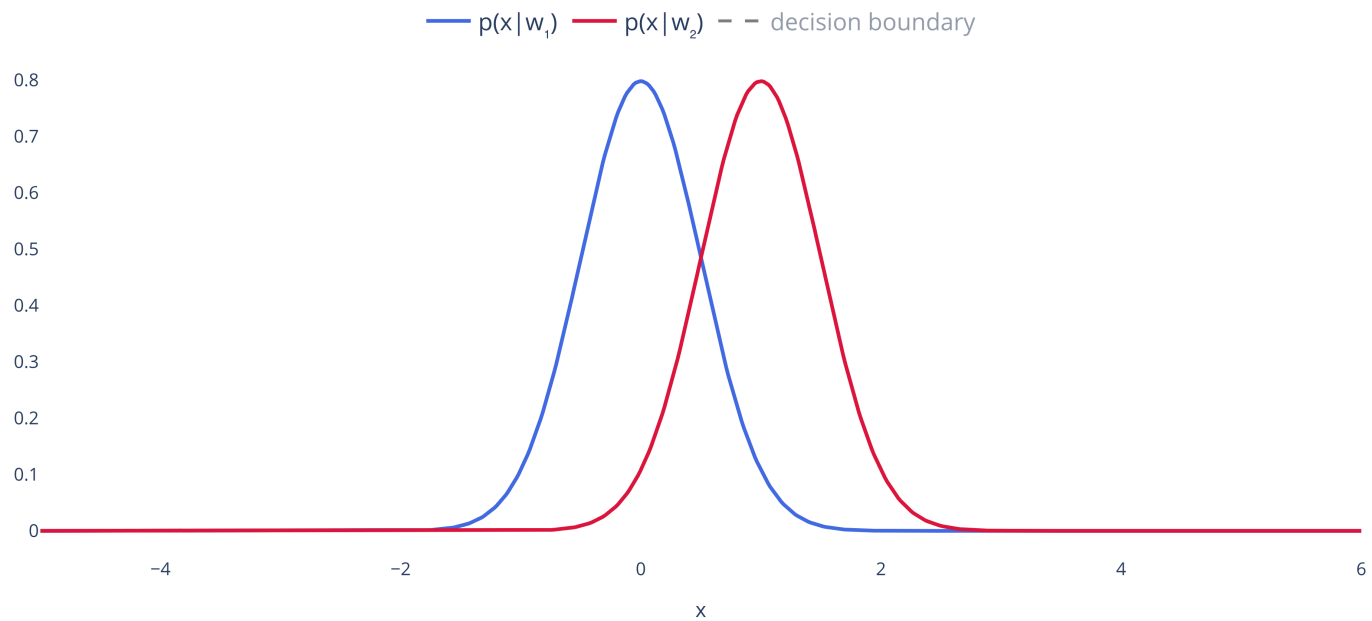


$\mu_1=0, \mu_2=1, \sigma_1=\sigma_2=0.5$

$\mu_1=0, \mu_2=2, \sigma_1=\sigma_2=0.5$

$\mu_1=0, \mu_2=1, \sigma_1=0.5, \sigma_2=0.6$

$\mu_1=0, \mu_2=1, \sigma=0.5, P(w_1)=0.1$



## Probability of error

- Want:
  - Algorithm that can solve task with high performance.
  - Think spam filter that removes all spam and keeps all ham.
- Problem:
  - Real data is complicated, we will make mistakes!
  - What is the probability of making an error with a Bayes classifier?
- The probability of error can be modelled as follows:

$$\begin{aligned} P_e &= P(\mathbf{x} \in R_1, w_2) + P(\mathbf{x} \in R_2, w_1) \\ &= \int_{R_1} p(\mathbf{x}|w_2)P(w_2) d\mathbf{x} + \int_{R_2} p(\mathbf{x}|w_1)P(w_1) d\mathbf{x} \end{aligned}$$

Bayes classifier minimizes  $P_e$

- Note that  $P_e$  can also be expressed in terms of an integral:

$$P_e = \int_{R_1} P(w_2)p(\mathbf{x}|w_2) d\mathbf{x} + \int_{R_2} P(w_1)p(\mathbf{x}|w_1) d\mathbf{x}$$

- Visual proof  $\rightarrow$
- Formal proof as extra exercise.

## Going beyond the two-class setting

- Given a new sample  $\mathbf{x}$ , how can we decide if we want to classify it as belonging to class  $w_i$ ?
- $P(w_i)p(\mathbf{x}|w_i) > P(w_j)p(\mathbf{x}|w_j)$  for all  $j \neq i$ .
- Assign  $\mathbf{x}$  to the largest  $P(w_i)p(\mathbf{x}|w_i)$ .

# Dealing with risk in the Bayes classifier

- In many cases, certain types of errors have bigger consequences than others.
- **Example 1:**
  - A spam email getting through the spam filter?
    - Annoying, but manageable.
  - A legitimate email with an important message getting caught in the filter?
    - Potentially very damaging.
- **Example 2:**
  - A benign tumor being classified as malignant in breast cancer screening?
    - Not good; increases workload and chance of unnecessary treatment. But, "better safe than sorry".
  - A malignant tumor being missed?
    - Potentially fatal.
- Can we incorporate this into the classifier?

## Introducing risk factors to the Bayes classifier

- Define:
  - $\lambda_{12}$ : penalty for wrongly classifying  $w_2$  as  $w_1$
  - $\lambda_{21}$ : penalty for wrongly classifying  $w_1$  as  $w_2$
- Consider the two-class setting again:
  - $\lambda_{12}P(w_1)p(\mathbf{x}|w_1)$   
 $> \lambda_{21}P(w_2)p(\mathbf{x}|w_2)$
- Rewrite as a ratio:

$$\frac{p(\mathbf{x}|w_1)}{p(\mathbf{x}|w_2)} > \frac{\lambda_{21}P(w_2)}{\lambda_{12}P(w_1)}$$

- Need to know class-conditional distributions.

## Investigating the simplest case — one-dimensional normal distributions

- Let  $p(\mathbf{x}|w_i)$  follow a normal probability density function.
- One-dimensional case:

$$p(x|w_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right)$$

- General (multivariate) case:

$$p(\mathbf{x}|w_i) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_i|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right)$$

## Example with two-class problem

- Consider a balanced dataset with class-conditional distributions  $p(\mathbf{x}|w_1) \sim \mathcal{N}(\mu_1, \sigma)$  and  $p(\mathbf{x}|w_2) \sim \mathcal{N}(\mu_2, \sigma)$ .
- Let  $P(w_1) = P(w_2)$  — balanced dataset (equiprobable classes).
- Expression for the ratio:

$$\frac{p(\mathbf{x}|w_1)}{p(\mathbf{x}|w_2)} > P(w_2)/P(w_1)$$



# Discriminant functions

- Take a step back:
  - What is actually going on?
- Input space is partitioned into regions. Two regions are separated by a decision surface.
- Often mathematically convenient to describe the decision surface using a *discriminant function*:
  - $g_i(\mathbf{x}) = f(P(w_i)p(\mathbf{x}|w_i))$

## Example for Gaussian case

- Revisiting the Gaussian likelihood.
- What could be a suitable choice for  $f$ ?
- In this case,  $g_i(\mathbf{x})$  becomes:

$$g_i(\mathbf{x})$$

## Decision surfaces

- Decision surface is where the discriminant functions are equal:
  - $g_{ij}(\mathbf{x}) = g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0$
- Quadratic surface! Draw example  $\rightarrow$
- Often we assume equal covariance structure. What happens then?

## Continue example for Gaussian case

- Remember:

$$g_{ij}(\mathbf{x})$$

$$= g_i(\mathbf{x})$$

$$- g_j(\mathbf{x})$$

$$= 0$$

- 1st term  $\rightarrow$
- 2nd term  $\rightarrow$

Special case for equiprobable classes with the same covariance matrix

- Assume  $\Sigma = \sigma \mathbf{I}$
- Then the discriminant function  $g_{ij}(\mathbf{x})$  becomes  $\rightarrow$
- Draw example  $\rightarrow$